

Exploring Dimensions of Human Likeness in Conversational AI: Implications for Human Identity Threat and Dehumanization

Yejin Lee¹ and Sang-Hwan Kim¹

Abstract

This study aims to identify humanlike traits in conversational AI (CAI) that influence human identity threat and dehumanization and to propose design guidelines to mitigate these effects. An online survey was conducted with 323 participants. Four key perceived dimensions of human likeness in CAI were identified through factor analysis: Self-likeness, Agency, Social Adaptability, and Memory & Communication. Structural equation modeling revealed that Self-likeness heightened both human identity threat and dehumanization, whereas Agency decreased dehumanization. Social Adaptability generally diminished the threat to human identity but intensified the effect of Self-likeness on human identity threat. Additionally, age negatively predicted both human identity threat and dehumanization, suggesting that younger individuals are more susceptible to these psychological effects. By uncovering the psychological structure behind users' perceptions of CAI anthropomorphism, this study deepens the understanding of the psychological and societal implications associated with humanlike CAI and informs its practical and ethical design.

Keywords

anthropomorphism, artificial intelligence, conversational AI, dehumanization, human-AI interaction, perceived human identity threat, structural equation modeling

Introduction

Conversational AI (CAI) refers to digital agents that communicate with users through natural language (Khatri et al., 2018). With recent advancements in AI, CAI has evolved beyond mere tools, now demonstrating humanlike cognition, emotions, and behaviors. Modern CAI systems recognize emotional cues, engage in social interactions, and even tell jokes, establishing a new paradigm of human-AI interaction that transcends purely instrumental relationships (Wieczorek, 2023).

Interacting with such humanlike CAI leads people to attribute human qualities to it—a phenomenon known as anthropomorphism (Epley et al., 2007). When CAI exhibits human characteristics, people naturally respond to it as they would in human interactions. However, the outcomes of AI anthropomorphism remain mixed (Blut et al., 2021). While some studies suggest that it fosters trust, enjoyment, and acceptance (e.g., Hsu & Lee, 2023; Pelau et al., 2021), others indicate that excessive anthropomorphism can threaten human identity (Li & Suh, 2021) and potentially lead to the dehumanization of actual humans (Kim & McGill, 2024).

This threat to human identity could lead to aggressive attitudes toward AI (Xi et al., 2024) and increased resistance to its use and adoption (Li & Suh, 2021). Moreover, dehumanization may cause individuals to view actual humans as interchangeable rather than unique (Dang & Liu, 2025). Prior research has raised concerns that such perceptions may lead to maladaptive outcomes in AI-integrated work environments, including diminished task engagement, increased stress, and mistreatment of human coworkers (Dang & Liu, 2025; Kim & McGill, 2024). Therefore, identifying the specific humanlike attributes of CAI that contribute to these adverse effects is essential for designing ethically responsible and psychologically sustainable AI systems. This aligns with recent research trends in anthropomorphism that emphasize the importance of distinguishing between

¹University of Michigan-Dearborn, USA

Corresponding Author:

Sang-Hwan Kim, Department of Industrial and Manufacturing Systems Engineering, University of Michigan-Dearborn, 4901 Evergreen Road, Dearborn, MI 48128, USA.
Email: dysart@umich.edu

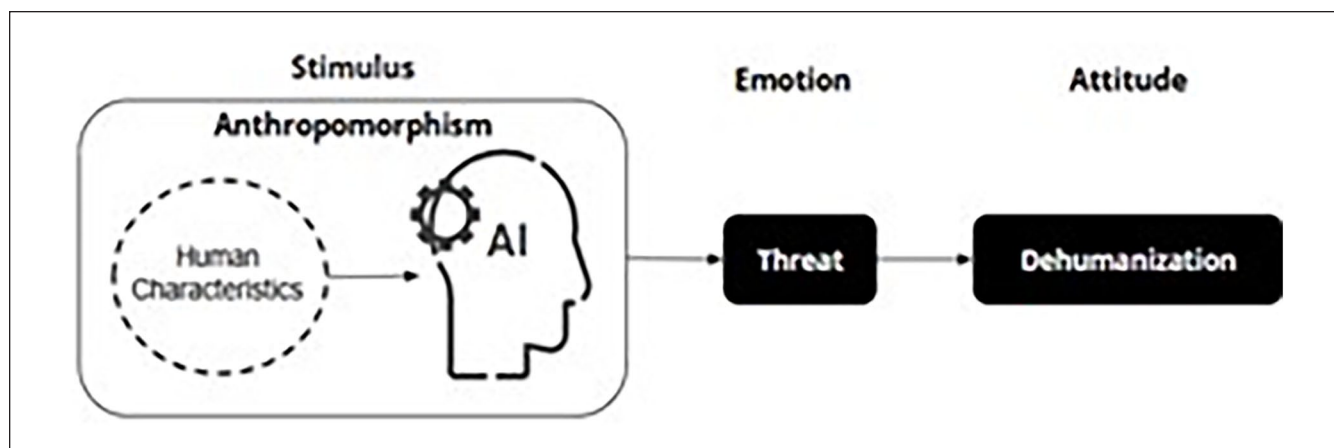


Figure 1. Conceptual framework.

qualitatively different dimensions of human likeness (Xi et al., 2024).

Accordingly, this study has two objectives: (1) to identify humanlike characteristics in CAI that influence human identity threat and dehumanization, and (2) to propose design guidelines to mitigate these effects. To achieve them, the study identifies the latent dimensions of perceived human likeness in CAI through factor analysis. It also examines the psychological structure and interrelationships among the perceived human likeness dimensions, human identity threat, and dehumanization using structural equation modeling (SEM). By revealing the psychological structure underlying users' perceptions of CAI anthropomorphism, this study provides theoretical and practical insights for designing CAI systems that balance the benefits of humanlike characteristics while preserving the human uniqueness of users. Figure 1 presents the conceptual framework of this study.

Literature Review

Frameworks to Define Humanness

"What does it mean to be human?" This question has long been explored by philosophers and psychologists, who have sought to identify human characteristics that distinguish humans from non-human entities.

Historically, the human mind has been regarded as the hallmark of humanness (Giger et al., 2024). To understand how we perceive the mind, Gray et al. (2007) proposed a mind perception framework that conceptualizes the mind along two key dimensions: *Agency*, which refers to the capacity for thought, planning, and self-control, and *Experience*, which encompasses the ability to feel emotions and sensations like pain, pleasure, and hunger. Their study showed that humans are perceived as having both Agency and Experience, while robots are viewed as having high Agency but low Experience. This difference underscores conscious experience as a crucial marker of humanness.

Expanding on this, Weisman et al. (2017) introduced the Body-Heart-Mind model, which includes three dimensions: *Body*, representing physiological sensations and biological needs; *Heart*, reflecting emotional depth and moral awareness; and *Mind*, encompassing cognitive abilities and goal-directed capacities. Their research showed that humans are perceived as high across all three dimensions, whereas robots are generally seen as high in Mind but low in both Body and Heart. This finding further highlights the importance of conscious experience in defining humanness.

Another theoretical model of humanness is Haslam's Dehumanization Model (Haslam, 2006), which distinguishes between two core types of traits: uniquely human (UH) and human nature (HN). UH traits, such as rationality, civility, refinement, and self-control, are socially acquired characteristics developed through culture and socialization. These traits are context-dependent and serve to differentiate humans from other animals. In contrast, HN traits, such as emotional responsiveness, interpersonal warmth, and individuality, are considered innate, universal, and biologically grounded. These qualities define the essential core of being human and distinguish humans from inanimate objects and machines. Emotionality, in particular, is regarded as central to humanness.

These frameworks consistently highlight that conscious experience, particularly emotional depth, is central to humanness, distinguishing humans from non-human entities.

Eroding the Boundaries: Threat to Human Identity and Dehumanization

What happens when non-human entities, such as CAI, begin to exhibit these human qualities? When CAI exhibits attributes fundamental to humanness, it may threaten human identity. According to Social Identity Theory (Tajfel & Turner, 1979), individuals are motivated to preserve the uniqueness of their ingroup—in this case, humanity—by distinguishing themselves from outgroups like machines. As a

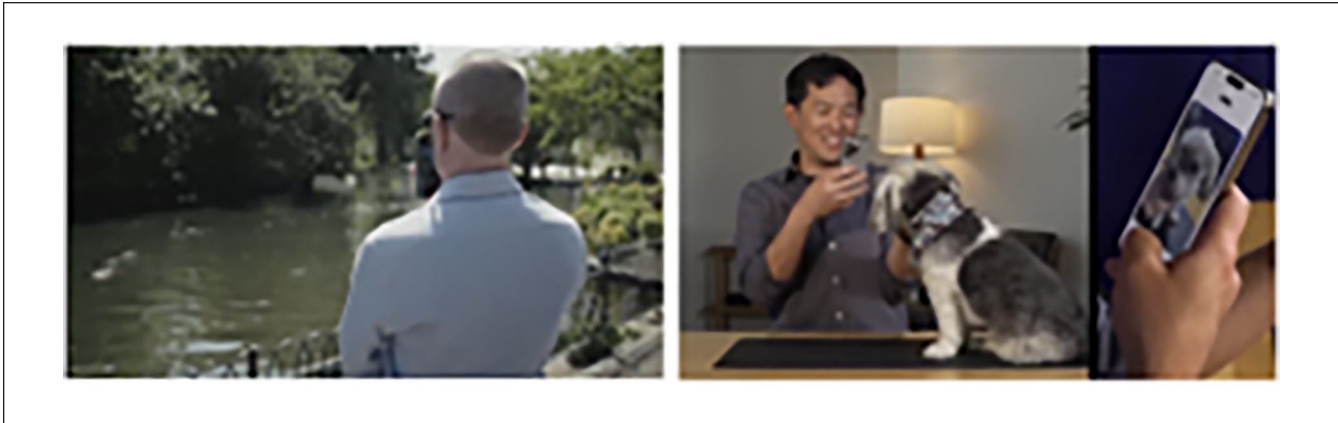


Figure 2. Screenshots from the video stimulus.

result, excessive perceived similarities between humans and CAI can raise concerns about blurred intergroup boundaries and evoke a sense of identity threat (Li & Suh, 2021; Złotowski et al., 2017).

Individuals may adopt defensive strategies to reaffirm human uniqueness and superiority in response to such threats. Research indicates that when human uniqueness in cognitive domains is challenged, people compensate by highlighting alternative attributes, such as emotional responsiveness, as core aspects of humanness (Cha et al., 2020). However, when AI systems encroach upon domains perceived as the essence of human uniqueness, people may feel forced to redefine what it means to be human, which can lead to feelings of inferiority and foster negative attitudes toward AI (Schaap et al., 2024).

This excessive blurring of the human-AI boundary may influence how individuals perceive genuine humanness. *Dehumanization* refers to denying human qualities to others, thereby reducing them to the status of objects or animals (Waytz et al., 2010). Recent research has identified a “dehumanization paradox,” in which attributing humanlike traits to AI paradoxically diminishes perceptions of human uniqueness (Chen et al., 2023). An empirical study by Kim and McGill (2024) supports this paradox: Participants who read articles and watched videos about socio-emotional AI reported greater dehumanization toward others compared to those exposed to AI with only cognitive abilities. In the socio-emotional condition, the participants consistently showed more negative treatment intentions toward other employees across three different scenarios and perceived less humanness in real humans. The study suggests that excessive similarity between AI and humans in the Experience dimension can lead to an assimilation effect, in which human qualities are perceived as increasingly comparable to those of AI, ultimately degrading perceptions of human uniqueness.

Dehumanization can occur both toward oneself and toward others. Prior research indicates that people generally

attribute more humanness to themselves, which makes them more resistant to self-dehumanization. In contrast, they are more likely to dehumanize others (Haslam & Bain, 2007; Park et al., 2016). However, when individuals perceive themselves as part of a devalued human group, they may also experience self-dehumanization.

To systematically examine both forms of dehumanization, separate SEM models are employed. The findings are expected to enhance the understanding of the psychological and societal implications of CAI anthropomorphism and to inform the practical and ethical design of CAI systems.

Methods

Initially, 35 key human characteristics were identified through a literature review of twelve studies in psychology, philosophy, and human-computer interaction (HCI) that focused on the concepts of humanness and anthropomorphism. In line with Epley’s (2018) viewpoint, which posits that true anthropomorphism involves attributing mental states and subtle humanlike qualities rather than merely physical resemblance, these 35 characteristics primarily reflect the psychological and cognitive traits repeatedly recognized as defining humanness.

Next, a brief passage was created to introduce the concept of CAI, along with a 1-minute and 33-second video that illustrates seven different interaction scenarios. These scenarios include CAI generating code for data visualization, creating images, describing a scene, responding affectionately to a user introducing their puppy, telling humorous jokes, offering appearance advice for a job interview, and conducting a mock interview. Sample screenshots from the video are shown in Figure 2.

Participants aged 18 or older, fluent in English, and residing in the United States were recruited from a university participant pool and Amazon Mechanical Turk (MTurk). Participants from MTurk were required to be Mechanical Turk Masters with a HIT approval rate above 85% and at

least 500 approved HITs. After obtaining informed consent, a total of 323 valid responses were collected. The sample consisted of 148 female and 172 male participants, with three individuals opting not to disclose their gender. The average age of the participants was 39.72 years ($SD=11.97$).

During the survey, participants read a passage and watched a video stimulus. They then rated the extent to which the CAI exhibited each of the 35 characteristics and assessed their perceived threat to human identity and uniqueness. Following this, dehumanization toward oneself and others was evaluated in random order, after which demographic questions were posed. All key variables were measured using a 7-point Likert scale, and two attention-check questions were included to ensure data quality. As compensation, three participants were selected through a random raffle to receive \$10.

For data analysis, factor analysis with oblimin rotation was conducted to identify the latent dimensions underlying the perceptions of the 35 characteristics. Partial Least Squares Structural Equation Modeling (PLS-SEM) analyses were performed using SmartPLS to examine the relationships among the perceived human likeness dimensions, perceived human identity threat, self-directed and other-directed dehumanization, and demographic information, including age and educational level.

Results

The factor analysis revealed four dimensions underlying the perception of the 35 human characteristics: Self-likeness, Agency, Social Adaptability, and Memory & Communication. This classification was based on eigenvalues greater than 1 and an analysis of the scree plot.

Self-likeness includes internal mental states and self-reflective traits, such as beliefs, maturity, and the capacity to experience complex emotions like remorse and love, which are fundamental aspects of what it means to be human. *Agency* encompasses rational thinking, judgment, planning, and execution. *Social Adaptability* reflects social skills, including empathy, humor, and relationship building, allowing CAI to respond appropriately in social situations and engage in interpersonal interactions. Lastly, *Memory & Communication* pertains to communication abilities and perceptual memory skills.

For the measurement model testing, we assessed reliability, convergent validity, and discriminant validity. Items with outer loadings below 0.708 were removed. The reliability coefficients (Cronbach's α and composite reliability) exceeded 0.70 but did not surpass 0.95, which aligns with the recommended thresholds (Hair et al., 2019). During this process, redundant items were removed to enhance construct validity. In addition, the average variance extracted (AVE) for each construct was greater than 0.50, indicating acceptable convergent validity (Fornell & Larcker, 1981). Discriminant validity was examined using both the Fornell–Larcker criterion and

Table 1. Retained Measurement Items by Construct.

Constructs	Items
Self-likeness	Experiencing remorse feelings
	Experiencing pain
	Feeling love
	Holding beliefs
	Having intentions
	Having culture
	Getting more mature
	Being creative and having its own imagination
Memory	Recognizing faces
	Remembering things
Social adaptability	Having a sense of humor
	Empathy
	Having relationships
	Getting better at coping with certain situations
Agency	Reasoning about things
	Planning
	Working toward a goal
	Making ethical decisions
	Technological advancements in the area of AI are threatening human uniqueness.
Perceived human identity threat	Recent advances in AI technology are challenging the very essence of what it means to be human.
	It threatens my very existence.
	It makes me worry about my place in the world.
	It makes people like me less important.
	It makes me worry about my own job security.
	It seems to lessen the value of my existence.
	I /They feel superficial like I /they have no depth.
	I am /They are mechanical and cold like a robot.
Dehumanization (self/others)	I /They lack self-restraint like an animal.
	I am /They are unsophisticated.

the Heterotrait–Monotrait (HTMT) ratio. All AVE values were greater than the squared correlations with other constructs, meeting the Fornell–Larcker criterion (Fornell & Larcker, 1981). Additionally, all HTMT values were below the threshold of 0.90, confirming adequate discriminant validity (Hair et al., 2019). Table 1 displays the items that were retained in the final models.

The overall fit of the models was evaluated using R^2 and SRMR. The R^2 values showed that the self-dehumanization model accounted for 32.4% of the variance in perceived human identity threat and 60.0% in self-directed dehumanization. Similarly, the other-dehumanization model explained 32.3% of the variance in perceived human identity threat and 58.3% in other-directed dehumanization. In addition, the SRMR was 0.069 for the self-dehumanization model and 0.070 for the other-dehumanization model, both below the recommended maximum of 0.10, indicating acceptable fit (Hair et al., 2013).

The path analysis demonstrated consistent patterns in both self-dehumanization and other-dehumanization models.

The Self-likeness dimension directly increased the perceived threat to human identity and dehumanization, while

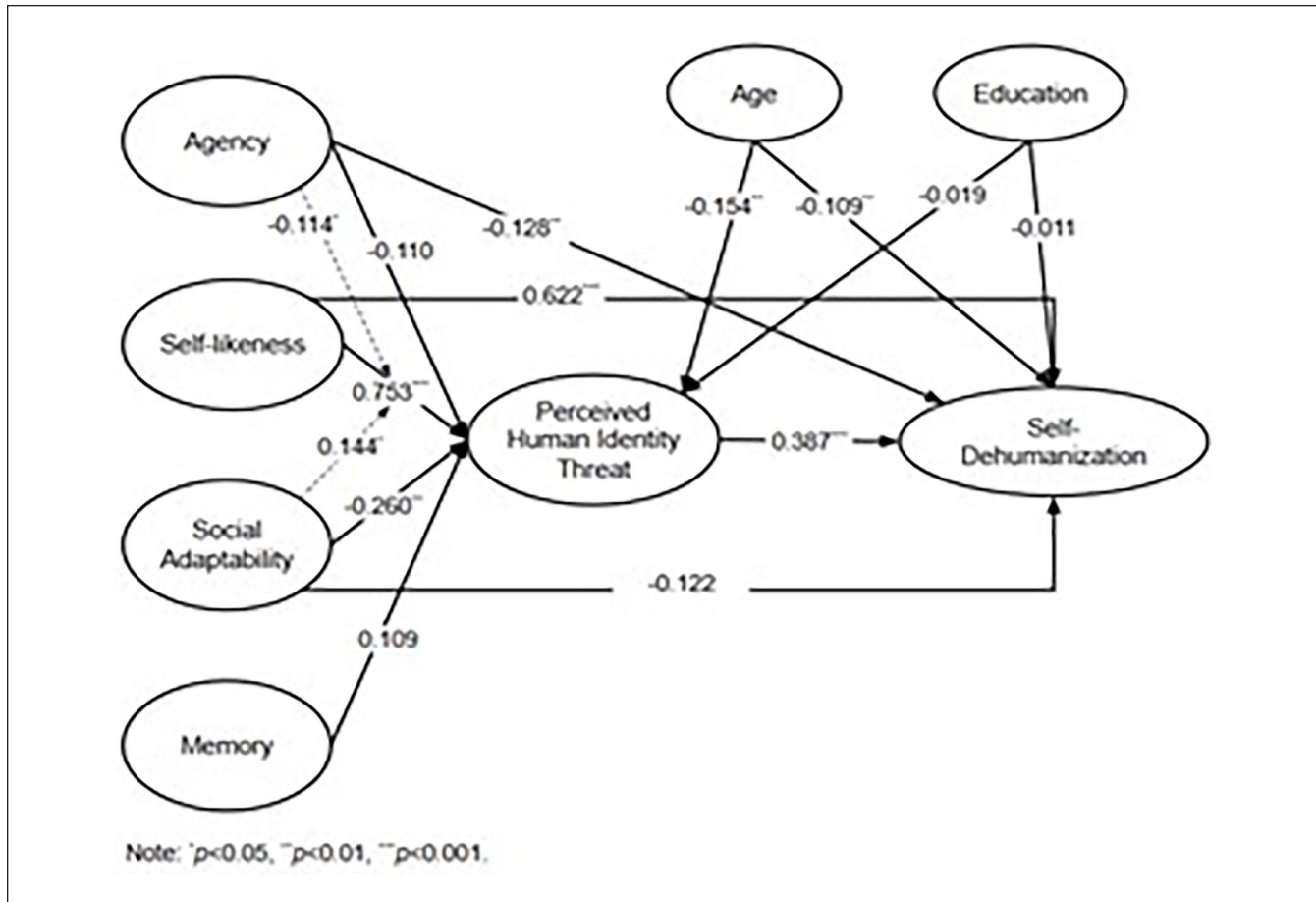


Figure 3. Self-dehumanization model.

also indirectly increasing dehumanization through the perceived threat to human identity. This suggests that when CAI is perceived as possessing mental capacities associated with self-conscious experience, such as beliefs, love, and maturity, the boundary between humans and machines becomes blurred. This blurred line may threaten human identity, leading individuals to attribute less humanness to actual humans.

In contrast, the Agency dimension significantly reduced self-directed dehumanization and marginally reduced other-directed dehumanization ($p = .054$). It also negatively moderated the effect of Self-likeness on perceived human identity threat. A moderated mediation analysis revealed that the interaction between Agency and Self-likeness had an indirect influence on dehumanization through perceived human identity threat. This suggests that when CAI is viewed as rational and goal-directed—traits traditionally associated with machines—it helps restore balance and mitigates the perceived threat to human identity.

The dimension of Social Adaptability reduced the perceived threat to human identity and, in turn, indirectly decreased dehumanization. However, it also intensified the influence of Self-likeness on perceived human identity threat. A moderated mediation analysis further revealed

that the interaction between Social Adaptability and Self-likeness indirectly increased dehumanization through perceived human identity threat. These findings suggest that when CAI is perceived as both highly self-like and socially skilled, it may seem excessively humanlike, thus intensifying the threat to human identity and ultimately promoting greater dehumanization.

Finally, age negatively predicted both perceived human identity threat and dehumanization. Younger individuals reported experiencing higher levels of threat and, as a result, greater dehumanization. Mediation analysis indicated that the perceived human identity threat significantly mediated the relationship between age and dehumanization. This suggests that the heightened sense of threat among younger individuals contributed to their increased tendency to dehumanize others.

Specific path coefficients are shown in Figures 3 and 4.

Conclusion

This study advances the understanding of CAI anthropomorphism and its psychological and societal implications by systematically identifying key humanlike characteristics in CAI

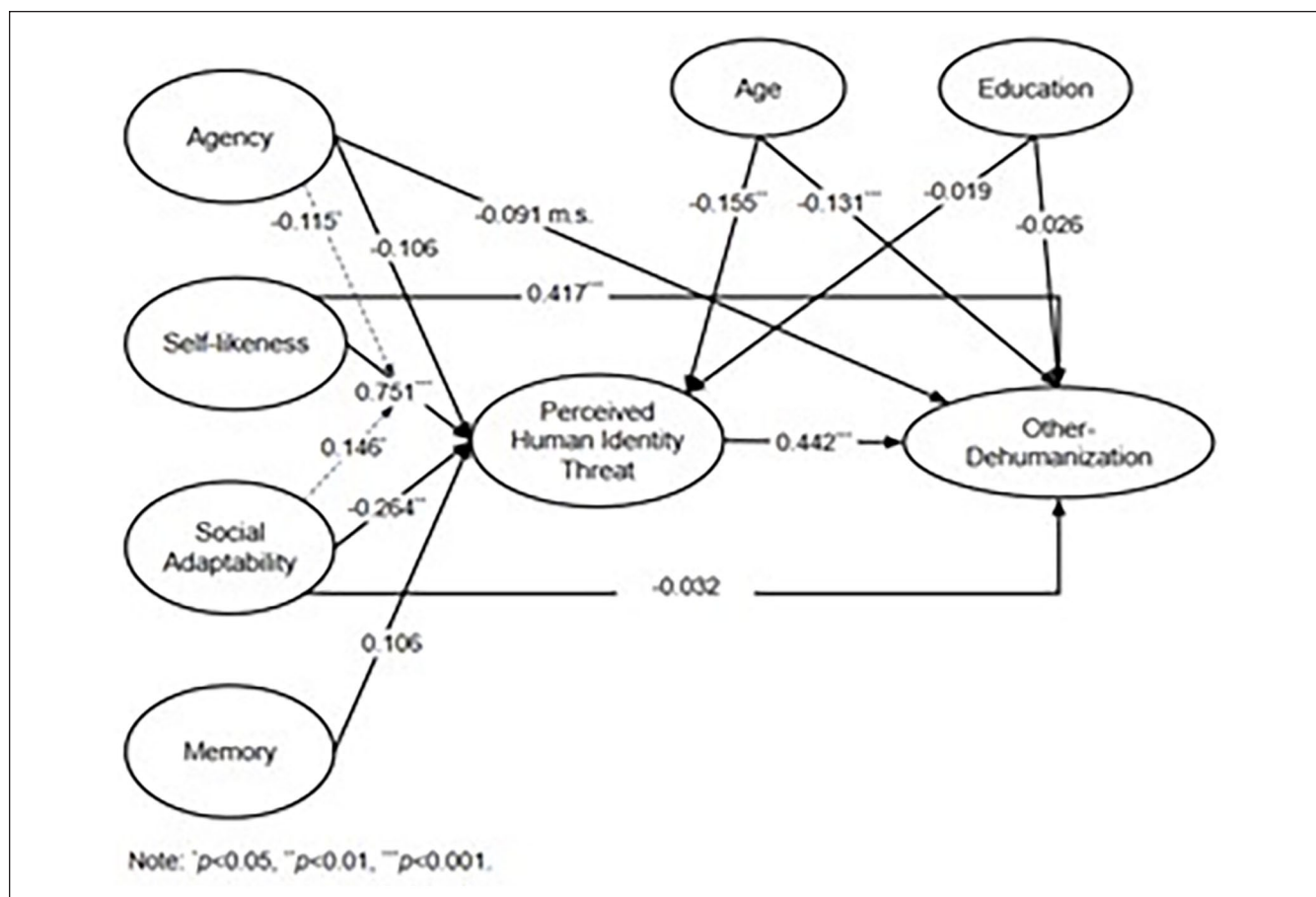


Figure 4. Other-dehumanization model.

and examining their effects on perceived human identity threat and dehumanization.

From an academic perspective, this study proposes a four-dimensional measurement framework of CAI anthropomorphism, developed through an interdisciplinary literature review and validated via factor analysis. It also empirically demonstrates the psychological mechanisms that link the humanlike traits of CAI to perceived human identity threat and dehumanization through SEM, thereby advancing the ongoing discourse on the consequences of anthropomorphizing non-human agents.

From a practical standpoint, the study offers design guidelines for ethically designing humanlike CAI. First, relational AI, such as social robots and chatbots, should foster connections through empathy and humor while avoiding expressions of deep emotions or beliefs. Second, in self-expressive domains like AI therapy or coaching, CAI should adopt goal-oriented and structured roles to balance its Self-likeness and Agency. Third, when designing CAI for younger users, such as AI tutors, it is important to limit emotional mimicry to prevent blurring the distinctions between humans and AI, which can lead to identity-related confusion. Lastly, designers should clearly communicate the non-human identity of

CAI through symbolic design elements and verbal cues (e.g., "I'm your assistant"). By distinguishing between internally directed emotional experiences (Self-likeness) and externally directed social skills (Social Adaptability), this study offers more nuanced design guidelines for CAI service industries where emotional interaction with AI plays a central role.

Moreover, the findings present significant practical implications for the evaluation of CAI. The four identified dimensions of perceived human likeness in CAI can serve as diagnostic criteria to determine if a specific CAI design may pose psychological risks, particularly concerning threats to human identity and user dehumanization. Designers can utilize these dimensions to assess whether certain features and functionalities reinforce perceptions of excessive similarity to humans or support psychologically safe interactions by maintaining a balanced perception.

Building on these insights, future research should incorporate behavioral and longitudinal measures to explore how intimate, long-term interactions with CAI systems, such as AI friends or companions, may lead individuals to perceive humans as more replaceable than unique, thereby influencing their perceptions of genuine human relationships.

Additionally, examining individual differences, such as prior AI experience, as potential moderators may further inform the ethical design of personalized CAI systems. It is also crucial to re-examine the influence of the Memory & Communication dimension using more comprehensive items to ensure the model's robust validity.

By integrating theoretical advancements with practical applications, this research provides a foundational framework for responsible AI development, guiding future work in both CAI ethics and human–CAI interaction design.

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ORCID iDs

Yejin Lee  <https://orcid.org/0009-0003-3911-4638>
Sang-Hwan Kim  <https://orcid.org/0000-0002-2748-5664>

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